2016-DSE
BIO
PAPER 1B

B

Please stick the barcode label here.

Candidate Number

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2016

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) Present your answers in paragraphs wherever appropriate.

(7) The diagrams in this section are **NOT** necessarily drawn to scale.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

A130E01B

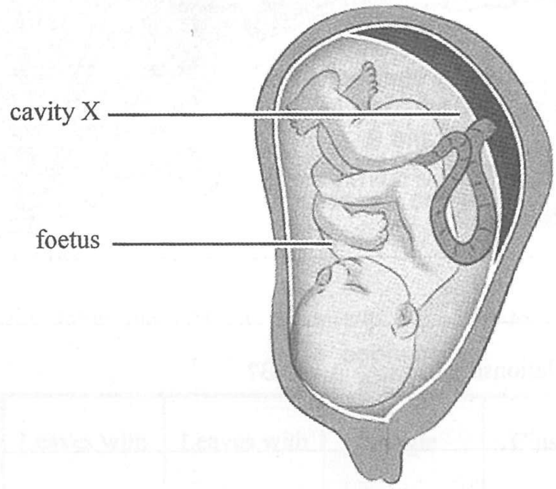
SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. The table below shows the condition which may result from damage to a certain part of the brain. Complete the table by filling in either the condition or the structural part of the brain affected. (3 marks)

Condition	***Structural part of the brain***
difficulty in breathing	
	cerebellum
difficulty in speech and vision	

2. The following diagram shows a foetus and the associated structures inside the uterus:



(a) What is the fluid found in cavity X? (1 mark)

....

(b) On the diagram, label the structure where the exchange of materials between the foetal blood and maternal blood takes place. (1 mark)

(c) Give ***two*** reasons why foetal blood has to be separated from maternal blood. (2 marks)

....

....

....

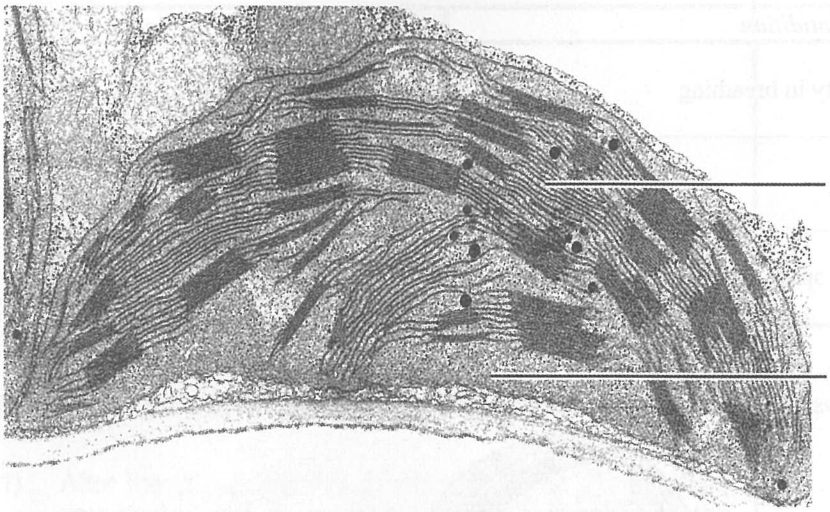
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

3. The diagram below shows the electron micrograph of an organelle:



A: ....

....

B: ....

(a) Label A and B. (2 marks)

(b) State a type of plant cell that contains this organelle. (1 mark)

(c) What is the functional relationship between A and B? (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

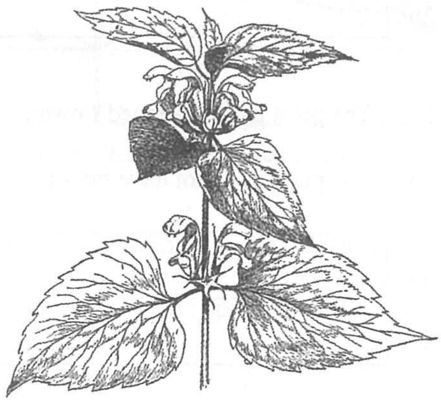
4. The following diagrams show the appearance of five flowering plants:

lesser celandine

hyacinth

wild daffodil

primrose



dead nettle

(a) In the following table, put a '✓' in the appropriate boxes to show the features of each flowering plant. (2 marks)

	Leaves with parallel veins	Leaves with network veins	Single flower	Cluster of flowers	Other features
lesser celandine					heart-shaped leaves
hyacinth					funnel-like flowers
wild daffodil					trumpet-like flowers
primrose					club-shaped leaves
dead nettle					two-lipped flowers

Please stick the barcode label here.

(b) Using the information from the table in (a), complete the following dichotomous key: (3 marks)

1a	The plant has leaves with parallel veins	....	2
1b	The plant has leaves with network veins	....	3
2a		....	hyacinth
2b		....	wild daffodil
3a	The plant has two-lipped flowers	....	
3b	The plant does not have two-lipped flowers	....	
4a		....	lesser celandine
4b		....	primrose

(c) Study the following statement:

The dichotomous key shows that the lesser celandine and primrose have a closer evolutionary relationship.

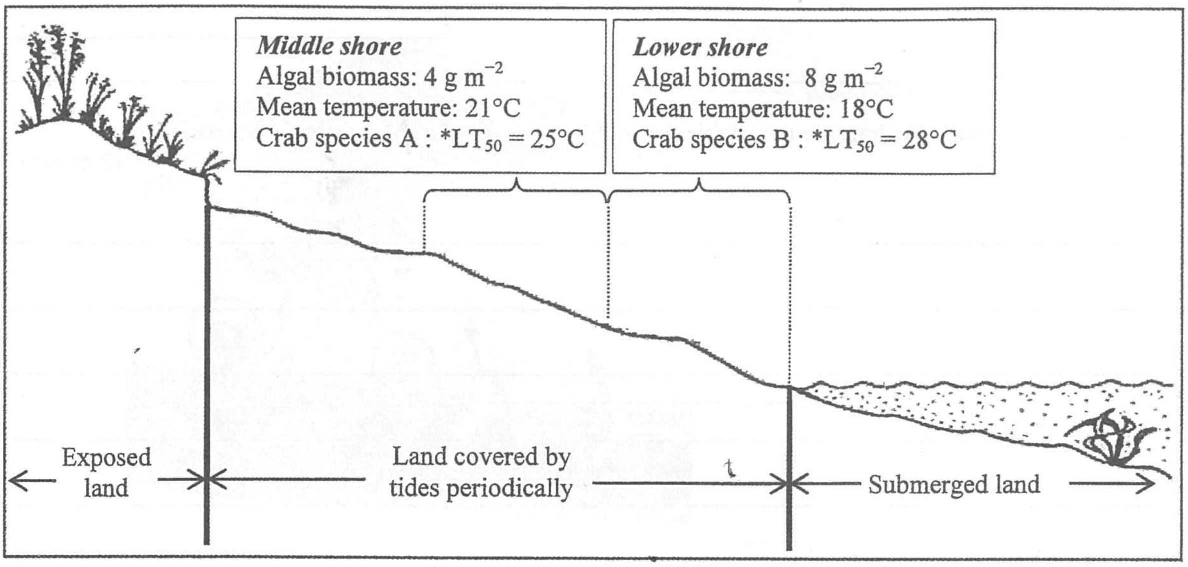
Do you agree? Explain your answer. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. The diagram below shows some biotic and abiotic factors of a rocky shore, the distribution of two crab species and their temperature tolerance:



* Median lethal temperature (LT_{50}): When a species is exposed to that temperature for 24 hours, half of the individuals die.

(a) The two species coexist on a rocky shore and feed on the same species of alga. When the two crab species are kept in a simulated habitat with the algal species, they will fight against each other. According to the information given in the diagram, deduce which crab species, A or B, would be a stronger competitor. (3 marks)

(b) Deduce whether temperature tolerance is a determining factor for the distribution of these crab species. (4 marks)

Please stick the barcode label here.

(c) Explain why quadrats are unsuitable for studying the abundance of crabs on the rocky shore. (2 marks)

Answers written in the margins will not be marked.

6. (a) Kitty and Karen are identical twins. Kitty has preferred meats to vegetables in her diet since her childhood. Kitty suffered from colon cancer at age 35 while Karen had the same disease 10 years later.

(i) Why did both sisters suffer from colon cancer? (1 mark)

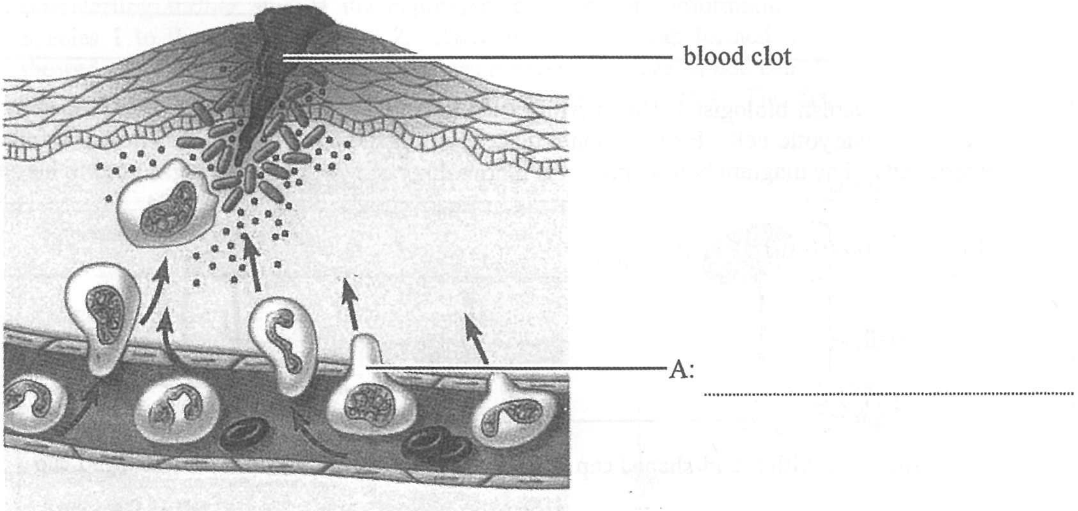
(ii) Why did the disease occur at different ages? (1 mark)

(b) Give *two* other lifestyles that increase the risk of suffering from cancer. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The diagram below shows a site of injured skin exhibiting an inflammatory response:



(a) Label the type of white blood cell represented by cell A in the above diagram. (1 mark)

(b) Explain why the tissue exhibiting the inflammatory response usually shows symptoms such as redness, swelling and pain. (3 marks)

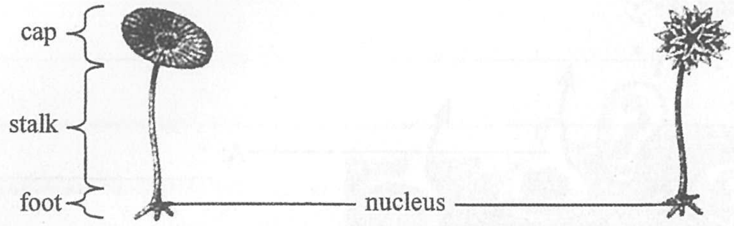
(c) Cell A will present the antigens of the invading pathogens to the lymphocytes. Describe what will happen subsequently. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

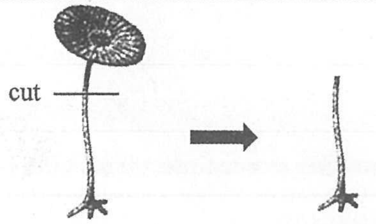
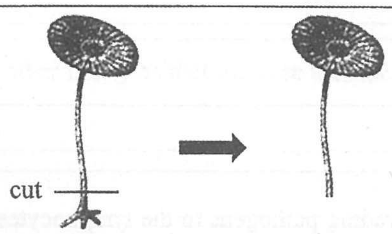
8. In 1930s, a Danish biologist J. Hämmerling tried to find out where the genetic information was stored inside the eukaryotic cell. He used some unicellular algae called *Acetabularia* to carry out a series of experiments. The diagram below shows the morphology of two species of algae used in his study:



Species 1 with round-shaped cap

Species 2 with flower-shaped cap

He divided Species 1 into two groups, removing the cap from one group (I) and the foot from another group (II). He then observed if any regeneration occurred in the remaining parts. The diagram below shows the treatments and the results:

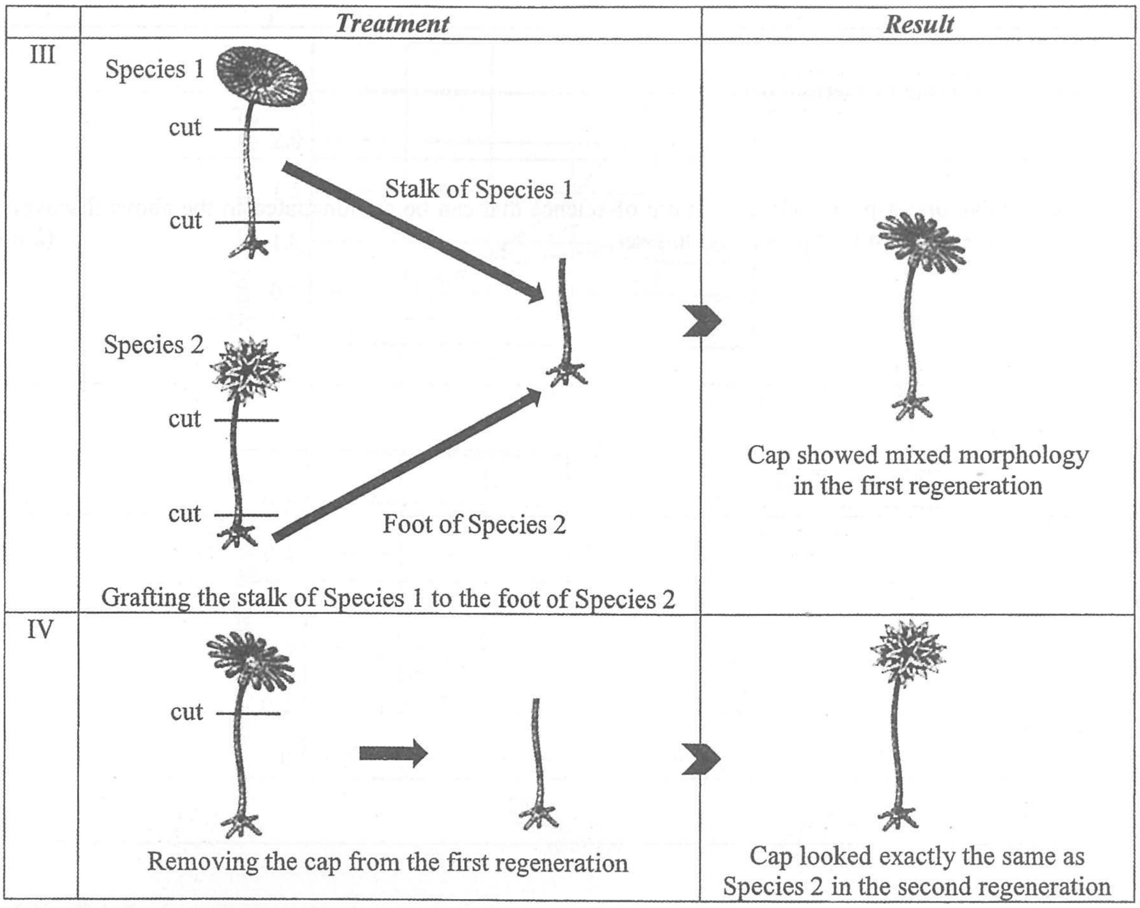
	Treatment	*Result*
I	 Removing the cap from Species 1	
II	 Removing the foot from Species 1	

(a) Describe the results of the above experiment. (2 marks)

(b) Based on the results, Hämmerling hypothesized that:

The hereditary information is stored in the foot of the algal cell.

Hämmerling further studied the expression of hereditary information by grafting the stalk of Species 1 to the foot of Species 2. After grafting, the cap formed from the first regeneration showed a mixed morphology (III). He then removed the regenerated cap. The cap formed from the second regeneration looked exactly the same as Species 2 (IV). The diagram below shows the treatments and the results:



(i) Hämmerling concluded that when the cut stalk was transplanted, it contained some short-lived instruction derived from the foot of Species 1, resulting in a mixed morphology of the cap in the first regeneration.

(1) Suggest the type of biomolecule that carried the short-lived instruction. (1 mark)

(2) How could the biomolecule suggested in (1) affect the morphology of the cap? (2 marks)

Answers written in the margins will not be marked.
Answers written in the margins will not be marked.

(ii) How do the results from this experiment support Hämmerling's hypothesis? (2 marks)

(c) Give *one* aspect about the nature of science that can be demonstrated in the above discovery and give a reason to support your answer. (2 marks)

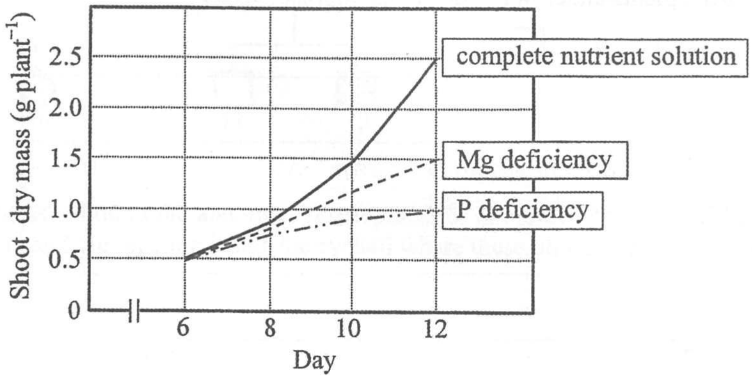
Answers written in the margins will not be marked.

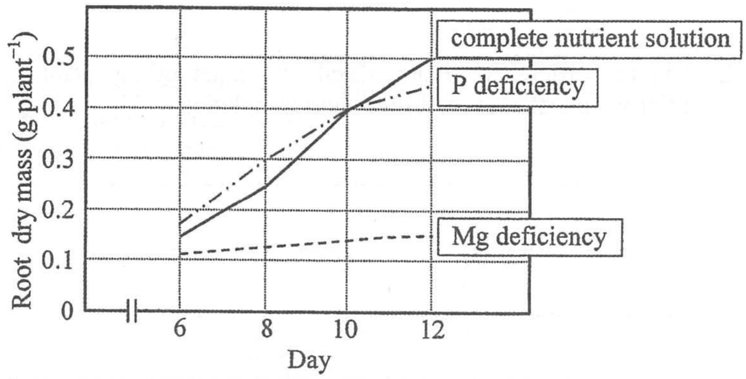
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. To study the effect of mineral deficiency on shoot and root dry masses, bean plants were grown in a complete nutrient solution (a solution containing all essential nutrients for growth) or a nutrient solution without either phosphorus (P) or magnesium (Mg) for 12 days respectively. The dry masses of shoot and root were then measured. The results are shown in the graphs below:

Key: ——— complete nutrient solution
—··—·· P deficiency
------- Mg deficiency





(a) Briefly describe how the dry mass of a plant can be determined. (2 marks)

....

....

....

....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(b) The leaves of the bean plants grown under Mg deficient conditions appeared yellow.

(i) Why did the leaves appear yellow? (1 mark)

(ii) Use this phenomenon to explain the results of the shoot dry mass and root dry mass of the bean plants under Mg deficient conditions. (3 marks)

(c) (i) Explain the difference in the overall dry mass of the plant grown under P deficient conditions and that in the complete nutrient solution. (2 marks)

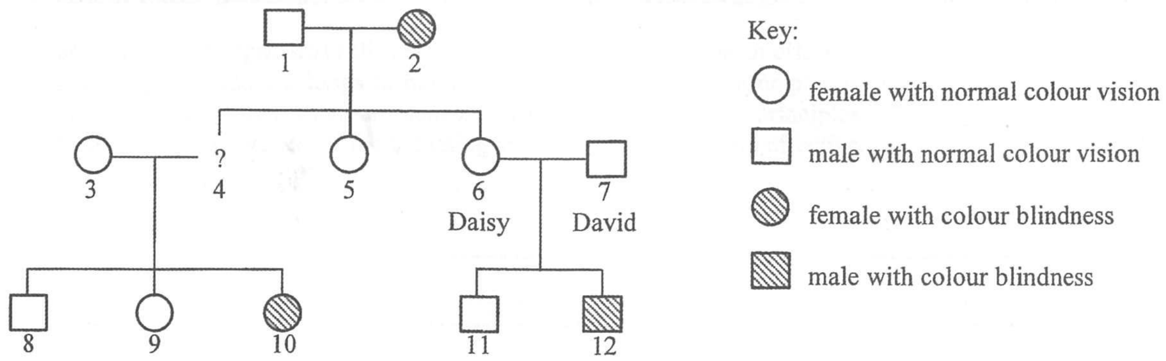
(ii) It was hypothesized that P inhibits the export of photosynthetic products from leaves to roots. Use this hypothesis to explain the results of the shoot dry mass and root dry mass of the bean plants under P deficient conditions. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. Colour blindness is an X-linked recessive genetic disorder. The pedigree below shows the inheritance of colour blindness in a family:



(a) Colour blindness is due to the abnormal development of photoreceptors. State the relevant type of photoreceptors and the location inside the eyeball where these photoreceptors are most abundant. (2 marks)

(b) Given that the dominant allele of colour vision is represented by R while the recessive allele is represented by r, determine all the possible genotypes and phenotypes of the offspring of individuals 1 and 2 using a genetic diagram. (5 marks)
(Note: Punnett square is not accepted)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(c) Draw all possible representation(s) for individual 4 with reference to the key of the pedigree. (1 mark)

(d) Daisy (individual 6) has recently given birth to a baby girl. Since one of her sons suffered from colour blindness (individual 12), Daisy worried that their daughter would have colour blindness too. David (individual 7) reassured her by saying that:

Don't worry. Our daughter will be fine because I have normal colour vision!

Justify David's claim. (5 marks)
(Note: Marks will not be awarded for genetic diagrams.)

For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Gas exchange in organisms relies very much on diffusion. To be an effective organ for gas exchange, the leaves in plants and the lungs in humans share some common principles in their structural adaptations. Discuss how their structures are adapted to fulfil these common principles. Despite these similarities, explain why the operation of the breathing system in human is more effective. (11 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

Answers written in the margins will not be marked.